Data Analytics 344

Assignment A1 — 2023

Rewritten Question Paper with New Model Answers

Compiled from the scanned examination script.

Note: Page 1 of the original document (cover / instructions page) was not included in the upload and is therefore not reproduced here. All other pages (2–21) were present and accounted for.

Section A: Customer Churn Prediction (SUNMobile) [20 marks]

Mobile operators focus on retention rather than acquisition since the cost of acquisition is typically five times higher than the cost of retention. The average monthly customer loss ratio (i.e. the percentage of customers that stop using a product in a given month) is approximately 4% for the mobile operator SUNMobile. To reduce the customer loss ratio, SUNMobile has hired a data analytics team to develop a model that can predict whether a customer is likely to leave SUNMobile in the future or not. The steps that the data analytics team proposed to create such a predictive model are illustrated in Figure 1 (Exploratory data analysis → Data Warehouse → Data pre-processing → Data extraction → Feature analysis → Sample data → Customer segmentation → Create predictive model).

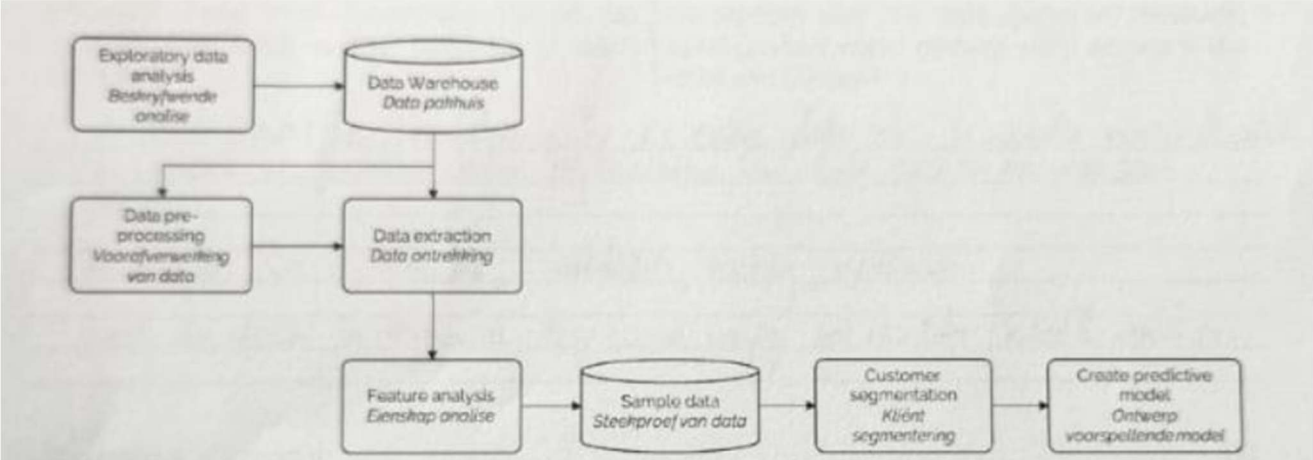


Figure 1: Proposed process to develop a predictive model
Figuur 1: Proses wat voorgestel is om 'n voorspellende model te ontwikkel

Question 1 *[3 marks]*

Figure 1 illustrates the data analytic process as a linear process and not a non-linear process. Is this depiction accurate? Support your answer with an example relevant to the case study.

Answer:

No, this is not fully accurate. In practice, the data analytics process is iterative rather than strictly linear — analysts frequently revisit earlier steps as new insights emerge.

For example, during feature analysis the team may discover that a feature needed for customer segmentation was dropped during data pre-processing (e.g. a usage-pattern variable), requiring them to return to the data extraction or pre-processing step before continuing. Similarly, if the predictive model performs poorly, the team may need to go back to feature analysis or customer segmentation to engineer new features.

Question 2 *[3 marks]*

Given that structured data is typically stored in a data warehouse, why is a data pre-processing step proposed?

Answer:

Data stored in a warehouse is structured for storage and general reporting purposes, not necessarily for analytics. It commonly contains missing values, inconsistent formats, duplicate records, outliers, and data spread across many normalised tables.

The pre-processing step is needed to clean, transform, and consolidate this raw warehouse data (e.g. handling missing values, correcting data types, removing duplicates, and joining relevant tables) into a form that is suitable as input to the modelling algorithms used later in the process.

Question 3 *[3 marks]*

Assuming that computational processing is unlimited, why is a sample of the data set created? How many samples should be created and what is the purpose of each sample?

Answer:

Even with unlimited computing power, a sample is still required because a model cannot be honestly evaluated on the same data it was trained on — doing so would give an overly optimistic view of its performance and hide overfitting. Held-out data that the model has not seen during training is needed to obtain a realistic estimate of how the model will perform on new, unseen customers.

Typically three samples are created: (1) a training set, used to fit/train the model; (2) a validation set, used to tune hyperparameters and compare candidate models during development; and (3) a test set, used only once, at the very end, to give an unbiased estimate of the final model's performance.

Question 4 *[3 marks]*

Is it more appropriate to use hold-out sampling or out-of-time sampling for the provided case study? Motivate your answer.

Answer:

Out-of-time sampling is more appropriate. Customer churn behaviour is influenced by time-varying factors (promotions, competitor actions, seasonality, network quality changes), so the relationship between features and churn can change over time (concept drift).

By training the model on data from an earlier period and testing it on data from a later period (out-of-time), SUNMobile can verify that the model generalises to future behaviour, which is exactly how the model will be used in production — predicting future churn from current data.

Question 5 *[2 marks]*

What data quality issue will occur based on the information provided in the description of the problem?

Answer:

Longevity / concept drift: because customer behaviour, market conditions, and competitor offerings change over time, historical patterns used to build the model may no longer accurately reflect current or future customer behaviour, reducing the model's ongoing validity and requiring periodic retraining.

Question 6 *[2 marks]*

The data analytics team advised that client segmentation should be performed to identify different customer loyalty groups before a predictive model is developed. How can the outcome of the data segmentation be used when developing the predictive model? Provide two different options.

Answer:

Option 1: Build a separate predictive model for each customer segment (e.g. high-loyalty vs low-loyalty customers), since the drivers of churn may differ substantially between groups, improving overall predictive accuracy.

Option 2: Use the segment/cluster label itself as an additional descriptive (input) feature in a single overall predictive model, allowing the model to learn segment-specific patterns without needing entirely separate models.

Question 7 *[4 marks]*

To implement customer segmentation the data analytics team used k-means clustering where the value of k was smaller than the number of instances in the data set. Upon analysing the clustering results the team found that one of the clusters contained no instances. Is it possible to have clusters with no instances? Explain your argument with an example.

Answer:

Yes, it is possible. K-means clustering will produce an empty cluster if the corresponding centroid ends up positioned such that no data point is closer to it than to any of the other centroids after an update step (this can happen with unlucky/poor initial centroid placement, especially with outlying initial centroids).

Example: suppose k = 3 and the data naturally forms two tight, well-separated groups. If one of the three initial centroids is randomly placed far from any data (e.g. in an empty region between or outside the two natural groups), then during the assignment step no points may ever be closer to that centroid than to the other two — leaving it with zero assigned instances.

Section B: Early Detection of Intrauterine Growth Restriction (IUGR) [26 marks]

Intrauterine growth restriction (IUGR) is one of the most common causes of stillbirths (i.e. death or loss of a baby before or during delivery). To help with the early detection of IUGR, a model can be developed that predicts whether the fetal weight will be below the 10th percentile at 34 to 37 weeks based on measurements taken at 20 to 24 weeks gestation. To build such a predictive model, two data sets were collected: the Week20 data set (4764 rows, 3 features: patientID, duration, beatslost — one-row-per-patient) and the Week34 data set (2767 rows, 3 features: patientID, IUGR3, IUGR10 — one-row-per-patient).

Table 1: Description of features
Tabel 1: Beskrywing van eienskappe

Data set *Datastel*	Feature *Eienskap*	Type *Tipe*	Description *Beskrywing*
Week20	patientID	Numerical *Numeries*	Unique patient identifier *Unieke pasiënt identifiseerder*
Week20	duration	Numerical *Numeries*	The total time (in seconds) that the heart rate of the fetus was recorded *Die totale tyd (in sekondes) wat die hartklop van die fetus opgeneem is*
Week20	beatslost	Numerical *Numeries*	The number of times irregular heartbeats were observed in the recording of the heart rate of the fetus *Die aantal here wat abnormale hartklop waargeneem was tydens die opname van die fetus se hartklop*
Week34	patientID	Numerical *Numeries*	Unique patient identifier *Unieke pasiënt identifiseerder*
Week34	IUGR3	Categorical *Kategories*	Whether or not the IUGR was less than 3% *Of die IUGR minder as 3% was of nie*
Week34	IUGR10	Categorical *Kategories*	Whether or not the IUGR was less than 10% *Of die IUGR minder as 10% was of nie*

Question 1 *[2 marks]*

To create an analytical base table, the data in the Week20 data set must be combined with the Week34 data set. You plan to use the left_join function in R. However, when executing your code you received the error: Could not find function left_join. What does the error message mean and what steps should you take to resolve the error?

Answer:

The error means R cannot find a function called left_join in any of the packages that are currently loaded — left_join() is not part of Base R, it is provided by the dplyr package (part of the tidyverse).

To resolve it: (1) install the package if it is not already installed, using install.packages("dplyr") (or install.packages("tidyverse")); and (2) load the package into the current R session with library(dplyr) (or library(tidyverse)) before calling left_join() again.

Question 2 *[2 marks]*

Assume that you fixed the error in the previous question and that the data sets have been loaded into R as data frames. What will the dimensions of the object returned by the R code left_join(Week34, Week20) be? Note any assumptions that you made.

Answer:

Assuming every patientID in Week34 also appears in Week20 (i.e. no unmatched keys, so no row duplication or NA-only rows are introduced) and that patientID is the join key, the result of left_join(Week34, Week20) keeps all rows of the left-hand data set (Week34) and adds the matching columns from Week20.

Dimensions: 2767 rows (all rows of Week34) by 5 columns — the 3 columns of Week34 (patientID, IUGR3, IUGR10) plus the 2 non-key columns of Week20 (duration, beatslost).

Question 3 *[2 marks]*

Should the feature IUGR3 be used as a descriptive feature to develop the predictive model? Motivate your answer.

Answer:

No, IUGR3 should not be used as a descriptive feature. IUGR3 (whether IUGR was less than 3%) is closely related to and effectively derived from the same underlying measurement as IUGR10 (whether IUGR was less than 10%), which is the target feature being predicted.

Including IUGR3 would cause target leakage — it would give the model information that is only available at the same time as (or that essentially reveals) the outcome being predicted, resulting in an unrealistically 'accurate' but practically useless model.

Before combining the two data sets, a data quality report was generated for each data set. The data quality report for the Week20 data set is provided in Table 2.

Voordat die twee datastelle gekombineer is, is 'n datakwaliteitverslag vir elkeen van die twee datastelle gegenereer. Die datakwalitietverslag vir die Week20-datastel word in Tabel 2 verskaf.

Table 2: Data quality report for the Week20 data set
Tabel 2: Datakwaliteitverslag vir die Week20-datastel

	Count	%Miss.	Card.	Min	1st Qrt.	Mean	Median	3rd Qrt.	Max	Std Dev.
patientID	4764	0.0	4764	1	1191.8	2382.5	2382.5	3573.3	4764	1375.4
duration	4764	0.0	770	0	1460	1528.6	1558.0	1660.0	3362.0	319.48
beatslost	4764	0.0	41	0	8.00	13.0	12.0	18.0	40.0	7.3

Question 4 *[2 marks]*

Consider the data quality report provided in Table 2 (Week20 data set: patientID, duration, beatslost — Count, %Miss, Cardinality, Min, 1st Qrt, Mean, Median, 3rd Qrt, Max, Std Dev). Does the table contain any potential missing values? Motivate your answer.

Answer:

The %Miss column shows 0.0% for all three features, so there are no explicitly missing (NA/blank) values recorded.

However, this does not rule out disguised missing values. For example, beatslost has a Min of 0 and duration also shows a Min of 0 — a duration of 0 seconds is not a plausible recorded heart-rate monitoring session, and this could represent a sentinel value used to encode a missing measurement rather than a true observation, so these should be investigated further.

Question 5 *[3 marks]*

The feature beatslost is correlated with the target feature and with the feature duration. What type of feature is the feature beatslost? Would you recommend removing the feature or not? Motivate your answer.

Answer:

beatslost is a redundant (correlated) descriptive feature, since it is correlated with another descriptive feature (duration) as well as with the target.

Recommendation: it is generally not necessary to remove it purely because of correlation with duration (redundancy between descriptive features does not itself harm most models, and tree-based/rule-based models handle correlated features reasonably well), but it should be reviewed for multicollinearity if a linear/logistic regression model is used. Since it is also correlated with the target, it likely carries genuine predictive value and should be kept unless multicollinearity is shown to materially hurt the chosen model.

Question 6 *[2 marks]*

With the goal in mind of building an accurate model, you decide to create a derived feature named my_feature. The feature my_feature is the ratio between the feature beatslost and the feature duration. How can you add the feature my_feature to the Week20 data set using dplyr functions? Provide the relevant R code.

Answer:

```
Week20 <- Week20 %>% mutate(my_feature = beatslost / duration)
```

Question 7 *[6 marks]*

To build a model that can predict the target feature IUGR10 you decided to implement a rule-based system. However, most of the descriptive features considered are numerical features. It is not advised to use numeric features in a rule-based system since it is very unlikely that the rule set will cover unseen data instances. The data mining system LERS uses hierarchical clustering to discretise numerical values into clusters before rule induction is applied. Illustrate how hierarchical clustering can be applied to create discretised groups by clustering the data provided in Table 3 (ID 1–8, Values 0.1, 0.12, 0.25, 0.5, 0.63, 0.7, 0.75, 0.9; Targets 1,1,1,1,0,0,0,0) into one cluster using the AGNES algorithm, single link and Manhattan distance. Complete Table 4 using the results obtained.

Table 3: Numerical data

Tabel 3: Numeriese data

ID *ID*	1	2	3	4	5	6	7	8
Value *Waarde*	0.1	0.12	0.25	0.5	0.63	0.7	0.75	0.9
Target *Teiken*	1	1	1	1	0	0	0	0

Table 4: Results of applying hierarchical clustering

Tabel 4: Resultate van die toepassing van hiërargiese groepering

Iteration *Iterasie*	Cluster assigned per iteration per ID *Groepering toegewys per iterasie per ID*							
	1	2	3	4	5	6	7	8
0	A	B	C	D	E	F	G	H
1	A	A	C	D	E	F	G	H
2	A	A	C	D	E	F	F	H
3	A	A	C	D	F	F	F	H
4	A	A	A	D	F	F	F	H
5	A	A	A	F	F	F	F	H
6	A	A	A	F	F	F	F	F
7	A	A	A	A	A	A	A	A

{This was the original student answer}

Answer:

Using AGNES with single-linkage and Manhattan (= absolute) distance on the sorted values 0.1, 0.12, 0.25, 0.5, 0.63, 0.7, 0.75, 0.9, the algorithm repeatedly merges the two closest clusters (single link = smallest distance between any pair of points in the two clusters):

Iteration 0: each ID is its own cluster: {1},{2},{3},{4},{5},{6},{7},{8}.

Iteration 1: merge IDs 1 & 2 (|0.1-0.12| = 0.02, the smallest gap) → {1,2},{3},{4},{5},{6},{7},{8}.

Iteration 2: merge IDs 6 & 7 (|0.7-0.75| = 0.05) → {1,2},{3},{4},{5},{6,7},{8}.

Iteration 3: merge {6,7} & 8 (|0.75-0.9| = 0.15, next smallest available) → {1,2},{3},{4},{5},{6,7,8}.

Iteration 4: merge {1,2} & 3 (|0.12-0.25| = 0.13) → {1,2,3},{4},{5},{6,7,8}.

Iteration 5: merge {1,2,3} & 4 (|0.25-0.5| = 0.25) → {1,2,3,4},{5},{6,7,8}.

Iteration 6: merge {5} & {6,7,8} (|0.63-0.7| = 0.07) → {1,2,3,4},{5,6,7,8}.

Iteration 7: merge {1,2,3,4} & {5,6,7,8} into a single cluster of all 8 instances.

(This sequence of merges is what should be used to populate the cluster-assignment columns of Table 4 for iterations 0 through 7.)

Question 8 *[2 marks]*

The clustering algorithm described in Question 7 should be stopped just before the accuracy that can be obtained when using a rule set will decrease as a result of merging clusters. Based on this statement and the result obtained in question 7, how many clusters should you discretise the given values into? Motivate your answer.

Answer:

Two clusters. Stopping at iteration 6 gives the two groups {1,2,3,4} and {5,6,7,8}, which correspond exactly to the two target classes (Target = 1 for IDs 1–4 and Target = 0 for IDs 5–8). This is the last merge that keeps the clusters perfectly aligned with the target labels — merging further (iteration 7) combines both classes into one cluster, which would definitely reduce rule accuracy, since a single discretised group can no longer distinguish the two target values.

Question 9 *[2 marks]*

Do you think that discretising numerical values is more likely to result in a model that overfits or underfits the data in the presence of target noise? Assume that the algorithm used to develop the model is PRISM.

Answer:

Underfits. Discretisation reduces a continuous feature to a small number of coarse groups/bins, which lowers the resolution/granularity of the descriptive information available to PRISM. In the presence of target noise, PRISM (which relies on discrete value-target combinations to induce rules) working with fewer, broader bins is more likely to miss fine-grained patterns and produce overly general rules — i.e. the model is more prone to underfitting rather than overfitting, since noise combined with coarse bins makes it harder, not easier, to fit spurious detail.

Question 10 *[3 marks]*

Analyse the data provided in Table 4 using the cut edge weight statistic (CEWS). Is instance four a noisy instance? Motivate your answer with reference to the CEWS algorithm.

Answer:

The cut edge weight statistic evaluates, for each instance, how strongly it is connected (in terms of similarity) to other instances within its own final cluster versus to instances in other clusters — a low CEWS value (weak connection to its own cluster) flags a potential noisy/outlier instance.

Instance 4 (value 0.5) lies at the boundary between the two natural clusters {1,2,3,4} and {5,6,7,8}, roughly equidistant from instance 3 (0.25) on one side and instance 5 (0.63) on the other. Its within-cluster edge weight (link to instance 3) is comparatively weak relative to a typical within-cluster link in this data set, suggesting instance 4 is a candidate noisy instance that sits awkwardly at the class boundary rather than being clearly representative of its assigned cluster.

Section C: Predictive Maintenance for Milling Machines (Company X) [26 marks]

Company X wants to develop a predictive model for milling machines with the goal of indirectly improving milling machine availability. To develop this predictive model, Company X collected 10 000 instances, each described by eight features: ProductID, Type, Temperature air, Temperature machine, Rotational speed, Torque, Tool wear, and Failed (binary target: TRUE/FALSE).

Table 5: Description of features
Tabel 5: Beskrywing van eienskappe

Feature *Eienskap*	Description *Beskrywing*
Productid	The unique machine identifier *Die unieke masjien identifiseerder*
Type	The type of the machine *Die tipe masjien*
Temperature air	The temperature of the air measured in degrees Celsius *Die tempratuur van lug gemeet in grade Celsius*
Temperature machine	The temperature of the machine measured in degrees Celsius *Die tempratuur van die masjien gemeet in grade Celsius*
Rotational speed	The speed of the machine was measured in Radians per second *Die spoed van die masjien wat in Radiale per sekonde gemeet is*
Torque	The torque of the machine measured in Newton meter *Die wringkrag van die masjien gemeet in Newton meter*
Tool wear	The usage of the current tool measured in minutes *Die gebruik van die hudige instrument gemeet in minute*
Failed	A binary feature indicating whether the milling machine failed (TRUE) or not (FALSE) *'n Binêre kenmerk wat aandui of the freesmajien gevaal het (WAAR) of nie gevaal het (ONWAAR).*

Question 1 *[3 marks]*

Consider the density plot of the descriptive feature Tool wear (Figure 2: density plots of Tool wear separately for Failed = FALSE and Failed = TRUE). Why is the plot inappropriate to establish if there is a relationship between the feature Tool wear and the target feature Failed?

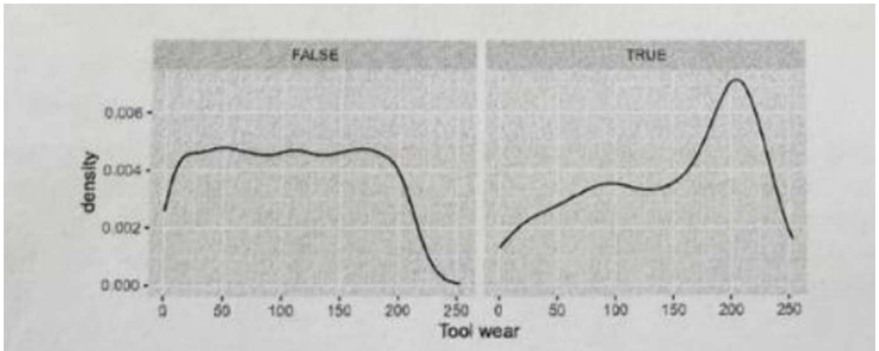


Figure 2: Density plot of Tool wear by Failure mode
Figuur 2: Digtheid plot van Tool wear by Failure tipe

Answer:

A density plot only shows the distribution/shape of Tool wear separately for each class — it does not show the actual joint frequency counts, and with a very imbalanced target (failures are rare relative to non-failures, as is typical in predictive maintenance) the two densities are each normalised to integrate to 1, which can visually exaggerate patterns in the much smaller Failed = TRUE group.

In addition, a density plot smooths the underlying data using a kernel and can be sensitive to the bandwidth chosen and to sparse/noisy regions, so it may show apparent structure (e.g. bumps) that do not reflect a genuine, statistically robust relationship. A more appropriate approach would be a statistical test (e.g. comparing means) or a binned frequency/proportion table accompanied by a significance test.

Question 2 *[3 marks]*

The tally of instances per Tool wear (binned into five groups) and Failed is provided in Table 6 (bins (0,50.6], (50.6,101], (101,152], (152,202], (202,253] with TRUE/FALSE counts: 28/1644, 42/1583, 34/1552, 51/1602, 69/395). Using Table 6, provide the rule set that will be learned if the OneR algorithm is applied.

Table 6: Observed frequencies
Tabel 6: Waargeneemde frekwensies

	Failed	
Tool wear	TRUE / *WAAR*	FALSE / ONWAAR
(0, 50.6]	28	1644
(50.6, 101]	42	1583
(101, 152]	34	1552
(152, 202]	51	1602
[202, 253)	69	395

Answer:

OneR selects, for the single feature under consideration (Tool wear), the majority class in each bin as the predicted rule outcome. For every bin the FALSE count is far larger than the TRUE count, so the majority class in all five bins is FALSE:

IF Tool wear IN (0, 50.6] THEN Failed = FALSE

IF Tool wear IN (50.6, 101] THEN Failed = FALSE

IF Tool wear IN (101, 152] THEN Failed = FALSE

IF Tool wear IN (152, 202] THEN Failed = FALSE

IF Tool wear IN (202, 253] THEN Failed = FALSE

Question 3 *[3 marks]*

Discuss any potential issues with the rule set obtained in Question 2. How can the training data be modified to obtain an improved rule set using the OneR algorithm? Assume that new data cannot be collected.

Answer:

Issue: because Failed = TRUE is a rare event, every bin's majority class is FALSE, so the resulting rule set always predicts Failed = FALSE regardless of Tool wear. This rule set has high overall accuracy (since most machines don't fail) but is completely useless for its purpose — it will never predict a failure, i.e. it has zero recall for the class of interest.

Fix without collecting new data: rebalance the existing training data, e.g. by undersampling the majority (FALSE) class and/or oversampling the minority (TRUE) class (or using a technique such as SMOTE on the existing records) so that the class distribution within each bin is less skewed. This gives OneR a realistic chance of learning bins where TRUE becomes the majority class, producing a more informative rule set.

Question 4 *[3 marks]*

The chi-squared test can be used to assess the association between two categorical features. Populate Table 7 with the expected frequency values between the features Tool wear and Failed.

Table 7: Expected frequencies
Tabel 7: Verwagte frekwensiewaardes

Tool wear	Failed: TRUE / WAAR	Failed: FALSE / ONWAAR
(0, 50.6]	16.3%	98.3%
(50.6, 101]	1.39%	91.4%
(101, 152]	2.14%	97.8%
(152, 202]	3.01%	96.9%
(202, 253]	14.9%	85.1%

{Original student answer, this seemed to have been wrong.}

Answer:

Expected frequency for each cell = (row total × column total) / grand total. Column totals: TRUE = 28+42+34+51+69 = 224; FALSE = 1644+1583+1552+1602+395 = 6776; Grand total = 7000. Row totals (bin totals): 1672, 1625, 1586, 1653, 464.

Expected TRUE per bin (row total × 224/7000): (0,50.6] ≈ 53.5; (50.6,101] ≈ 52.0; (101,152] ≈ 50.8; (152,202] ≈ 52.9; (202,253] ≈ 14.8.

Expected FALSE per bin (row total × 6776/7000): (0,50.6] ≈ 1618.5; (50.6,101] ≈ 1573.0; (101,152] ≈ 1535.2; (152,202] ≈ 1600.1; (202,253] ≈ 449.2.

Question 5 *[3 marks]*

How can the expected frequencies be calculated and presented in a table like Table 6 using functions from the tidyverse package? Provide the appropriate R code. Assume that you have a data set with the columns Tool wear (binned) and Failed, where each instance represents one measurement.

Answer:

```
library(dplyr); library(tidyr)

observed <- data %>% count(`Tool wear`, Failed)

totals <- observed %>% group_by(`Tool wear`) %>% mutate(row_total = sum(n)) %>% ungroup() %>% mutate(col_total = ave(n, Failed, FUN = sum), grand_total = sum(n))

expected <- totals %>% mutate(expected = row_total * col_total / grand_total) %>% select(`Tool wear`, Failed, expected) %>% pivot_wider(names_from = Failed, values_from = expected)
```

Question 6 *[1 mark]*

Provided that the chi-square statistic is large and the p-value is 2.16e-47, the null hypothesis should be _____?

Answer:

Rejected — the very small p-value (far below any conventional significance level such as 0.05) provides strong evidence against the null hypothesis of independence, indicating a statistically significant association between Tool wear and Failed.

Question 7 *[3 marks]*

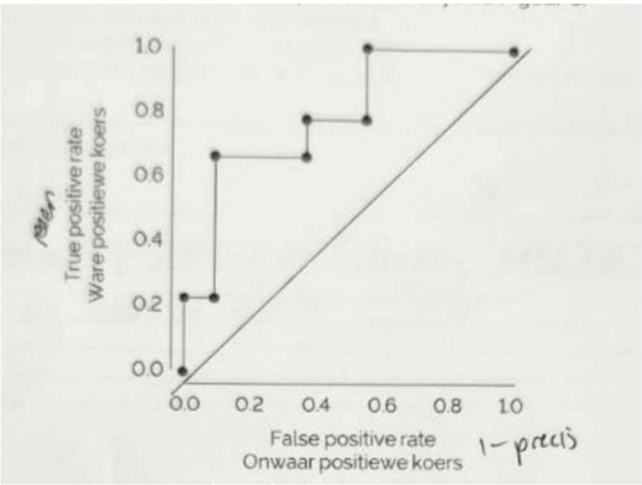
When selecting a performance measure to evaluate the predictive model, the company advised you that any potential failures could potentially injure machine operators. To prioritise safety above downtime, which classification metric should you use? Motivate your answer.

Answer:

Recall (sensitivity) = TP / (TP + FN) should be used. Prioritising safety means the company wants to minimise the number of actual failures that the model misses (false negatives), even if this means tolerating more false positives (unnecessary maintenance stops, i.e. downtime). Recall specifically measures the proportion of true failures correctly identified, making it the appropriate metric when missing a failure has serious safety consequences.

Question 8 *[4 marks]*

Consider the receiver operating characteristic (ROC) curve provided in Figure 3. Explain how the curve was generated and how the curve can be used to find an appropriate balance between safety and down-time (by referring to Figure 3).



Answer:

The ROC curve is generated by varying the classification (probability) threshold used to convert the model's predicted probabilities into class predictions, and at each threshold plotting the resulting True Positive Rate (recall) against the False Positive Rate. Each point on the curve corresponds to one threshold value; the curve as a whole shows the trade-off achievable across all possible thresholds.

To balance safety and downtime, the operator can pick a point (threshold) on the curve: moving toward the top-right of the curve increases the true positive rate (fewer missed failures → improved safety) but also increases the false positive rate (more unnecessary shutdowns → more downtime), while moving toward the bottom-left reduces false alarms/downtime at the cost of missing more real failures. Since safety has been prioritised, a threshold should be chosen that achieves a high true positive rate (accepting a higher false positive rate / more downtime) rather than optimising for overall accuracy alone.

Question 9 *[3 marks]*

Company X also requested that the speed of the model developed be considered. More specifically an accurate model that uses fewer features should be preferred. To find an accurate model that uses fewer features, Algorithm 1 is proposed (Algorithm 1: for every unique combination of features, train a model on the training set, measure performance on the training set, identify all models with sufficient performance, then select the model that uses the fewest features from those identified). What method of feature selection does Algorithm 1 apply? Are there any methodological mistakes in Algorithm 1?

Answer:

Algorithm 1 applies a wrapper method of feature selection (specifically an exhaustive/best-subset search), since it trains and evaluates a separate model for every possible combination of features and selects a subset based on model performance.

Methodological mistake: Step 3 measures performance on the same training data set that was used to train the model in Step 2. Evaluating performance on the training data gives an overly optimistic, biased estimate and will favour models/feature subsets that overfit, rather than ones that generalise well. Performance should instead be measured on a separate validation (held-out) set that was not used for training.

Section D: Predicting Wheat Yield [18 marks]

A farmer is interested in developing a model that can predict wheat yield close to the harvest period. The farmer has collected a data set with features: Longitude, Latitude, Year, Rainfall, Soil (ordinal, 1-5), NDVI, GNDVI, GSAVI, SIPI2 (all derived from aerial imagery) and Yield (kg/hectare, the target).

Table 8: Description of features
Tabel 8: Beskrywing van eienskappe

Feature *Eienskap*	Type *Tipe*	Description *Beskrywing*
Longitude	Numerical *Numeries*	Longitude of the plot *Lengtegraad van die plot*
Latitude	Numerical *Numeries*	Latitude of the plot *Breedtegraad van die plot*
Year	Numerical *Numeries*	Year that the measurements were taken *Die jaar wat die metings geneem is*
Rainfall	Numerical *Numeries*	Accumulative rainfall received from the planting date until the date of prediction *Akkumulatiewe reënval ontvang vanaf die plantdatum tot die datum van voorspelling*
Soil	Ordinal *Ordinaal*	Soil quality rating, ranging from 1 (poor soil) to 5 (good soil) *Grondkwaliteitgradering, wat wissel van 1 (swak grond) tot 5 (goeie grond)*
NDVI	Numerical *Numeries*	Normalised difference vegetation index derived from an aerial image *Genormaliseerde verskil plantegroei-indeks afgelei van 'n lugbeeld*
GNDVI	Numerical *Numeries*	Green normalised difference vegetation index derived from an aerial image *Groen genormaliseerde verskil plantegroei-indeks afgelei van 'n lugbeeld*
GSAVI	Numerical *Numeries*	Green soil-adjusted vegetation index derived from an aerial image *Groen grond-aangepaste plantegroei-indeks afgelei van 'n lugbeeld*
SIPI2	Numerical *Numeries*	Structure insensitive pigment index 2 derived from an aerial image *Struktuuronsensitiewe pigmentindeks 2 afgelei van 'n lugfoto*
Yield	Numerical *Numeries*	Wheat yield in kg/hectare *Koringopbrengs in kg/hektaar*

Table 9: Data quality report
Tabel 9: Datakwaliteitverslag

	Count	%Miss.	Card.	Min	1st Qrt.	Mean	Median	3rd Qrt.	Max	Std Dev.
Longitude	1183	0	1183	18.89	18.89	18.89	18.89	18.89	18.89	4e-04
Latitude	1183	0	1183	-33.9	-33.9	-33.9	-33.9	-33.9	-33.9	2e-04
Year	1183	0	2	2020	2020	2020.7	2021	2021	2021	0.5
Rainfall	1183	0	2	402	402	425	402	473	473	33.3
Soil	1183	0	5	1	2	3.2	3	4	5	1.3
NDVI	1183	0	905	0.45	0.79	0.82	0.85	0.88	0.92	8e-02
GNDVI	1183	0	899	0.48	0.68	0.71	0.73	0.76	0.81	6e-02
GSAVI	1183	0	968	0.22	0.50	0.55	0.56	0.60	0.70	8e-02
SIPI2	1183	0	464	0.87	0.91	0.92	0.92	0.04	1.09	1e-02
Yield	1183	0	1127	0.4	2297	3120.5	3132.4	3819.6	6356	1161

Question 1 *[2 marks]*

Since you do not understand what good crop yield values are, you are unsure how to determine if the model that you plan to develop performs adequately. What performance measure can you use to determine how your model performs in a unitless manner?

Answer:

A unitless (relative) performance measure such as the coefficient of determination (R^2), or a normalised error measure such as the Mean Absolute Percentage Error (MAPE) or a normalised RMSE (RMSE divided by the mean or range of Yield), can be used. These measures express model performance relative to the variability of the target itself (or as a percentage), rather than in the original kg/hectare units, allowing performance to be judged without needing domain knowledge of what a 'good' absolute yield value is.

Question 2 *[2 marks]*

When training a rule-based model on the data set you found that the rule set tends to exclude rainfall. Why is rainfall excluded, despite its practical importance in crop yield? Motivate your answer using relevant information contained in Table 9.

Answer:

According to Table 9, Rainfall has a relatively narrow spread in this data set (e.g. its Std Dev of 33.3 is small relative to its mean of ~402, and its 1st–3rd quartile range of roughly 402–473 is fairly tight), meaning that in this particular sample rainfall varies little across instances.

Because a rule-based algorithm looks for features whose values best split/discriminate the target, a feature with little variation across the data set carries little discriminatory power (it cannot meaningfully separate high-yield from low-yield instances even if it is practically important), so it tends to be excluded in favour of features that vary more across the observed data.

Question 3 *[3 marks]*

Figure 4 shows the relationship between the descriptive feature SIPI2 and the target feature Yield (a scatter plot showing a roughly diffuse, non-linear spread with no clear increasing/decreasing linear trend). What do you expect the Pearson correlation value to be? Explain how you derived the answer.

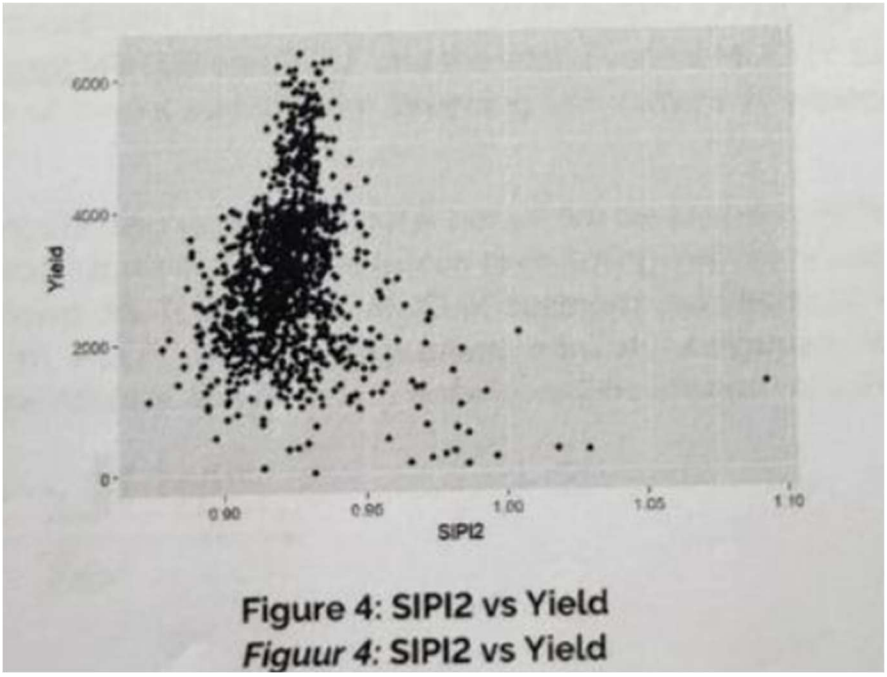


Figure 4: SIPI2 vs Yield
Figuur 4: SIPI2 vs Yield

Answer:

I would expect a Pearson correlation value close to zero, roughly in the range of about -0.1 to 0.1 (i.e. very weak).

This is because the scatter plot shows no clear linear trend between SIPI2 and Yield — the points are scattered broadly across the SIPI2 range at all yield levels rather than following a rising or falling straight-line pattern. Since the Pearson correlation coefficient specifically measures the strength of a linear relationship, a plot with a diffuse, non-linear (or non-existent) relationship will produce a value near zero, even if some other, non-linear association exists between the two variables.

Question 4 *[3 marks]*

You expect that values for the Soil feature will not always be available. To identify a substitute feature, you decided to calculate and cluster the Pearson correlation between features. Interpret the dendrogram provided in Figure 5 to determine which feature can potentially be used if the Soil feature is unavailable. Motivate your answer by explaining what the provided dendrogram illustrates.

Answer:

The dendrogram clusters features according to how correlated they are with one another (features that merge together at a low height are highly correlated; features that only merge at a high height are weakly related). In Figure 5, Soil merges directly with Longitude at a very low height, before joining any other feature group.

This indicates that Longitude is the feature most strongly (linearly) correlated with Soil in this data set, and therefore Longitude is the best candidate substitute feature to use if the Soil measurement is unavailable for a given plot.

Question 5 *[5 marks]*

How can we cluster the breeding lines into meaningful groups? Discuss how you can measure the similarity between two breeding lines. Assume hierarchical clustering will be applied. Hint: the breeding line MUTUS/ROLF//MUCUY is closer to the breeding line KUTZ//KFA/ROLF than it is to the breeding line KENYA/KACHU/KIDEA since it includes one of the wheat variants.

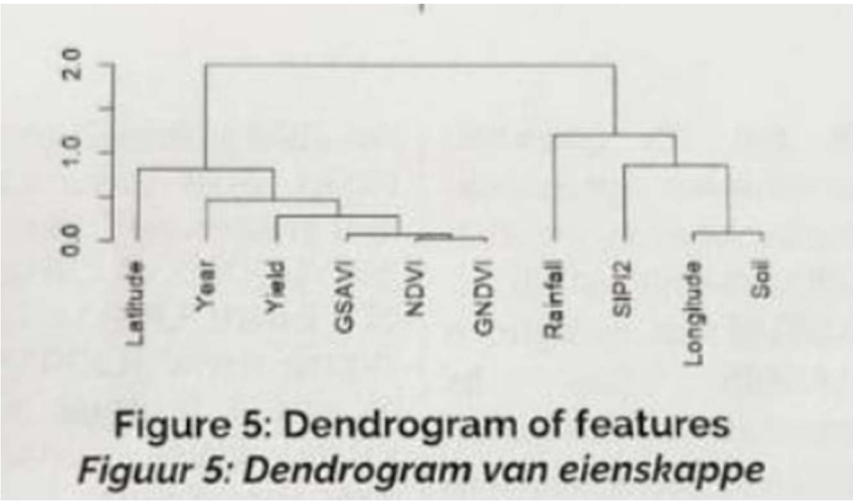


Figure 5: Dendrogram of features
Figuur 5: Dendrogram van eienskappe

Answer:

Breeding lines can be clustered hierarchically (e.g. using AGNES with an appropriate linkage method) based on a custom similarity/distance measure defined over the sets of wheat variants that make up each breeding line, rather than over numeric coordinates.

Similarity between two breeding lines can be measured using a set-overlap measure such as the Jaccard similarity coefficient: treat each breeding line as a set of its constituent wheat variants (e.g. MUTUS/ROLF//MUCUY = {MUTUS, ROLF, MUCUY}), and compute similarity as the size of the intersection of the two variant sets divided by the size of their union. Breeding lines that share more common parent variants will have a higher Jaccard similarity (and lower distance), which matches the given hint — MUTUS/ROLF//MUCUY shares the variant ROLF with KUTZ//KFA/ROLF, giving a non-zero similarity, whereas it shares no variants with KENYA/KACHU/KIDEA, giving zero similarity.

Question 6 *[3 marks]*

Consider the fact that some breeding lines will include multiple variants while other might only consist of two variants. This means that the breeding line MUTUS/ROLF//MUCUY will be as similar to KUTZ//KFA/ROLF then it is to KENYA/KACHU/KIDEA/ROLF when simply matching variant names. Suggest a way to calculate the similarity measure to account for the number of variants? In other words your measure should mathematically express that MUTUS/ROLF//MUCUY is more similar to KUTZ//KFA/ROLF than it is to KENYA/KACHU/KIDEA/ROLF.

Answer:

Use the Jaccard similarity coefficient instead of a simple shared-variant count: Similarity(A, B) = $|A \cap B| / |A \cup B|$, where A and B are the sets of variants in each breeding line.

This normalises the number of shared variants by the total number of distinct variants involved. For MUTUS/ROLF//MUCUY = {MUTUS, ROLF, MUCUY} vs KUTZ//KFA/ROLF = {KUTZ, KFA, ROLF}: intersection = {ROLF} (size 1), union size = 5, so Jaccard = 1/5 = 0.2. For MUTUS/ROLF//MUCUY vs KENYA/KACHU/KIDEA/ROLF = {KENYA, KACHU, KIDEA, ROLF}: intersection = {ROLF} (size 1), union size = 6, so Jaccard = 1/6 ≈ 0.167. Because the union (denominator) grows with the total number of variants involved, the measure correctly gives a higher similarity to the breeding-line pair with fewer total variants, as required.

Section E: Recognising Human Activity from a Wearable Accelerometer [10 marks]

A small data set was collected to recognise human activity from a wearable accelerometer sensor. Table 7 contains 8 instances with the average acceleration in the X, Y and Z directions within a short time window. There is no target feature; instead clustering will be used to determine if there are meaningful groups. Data: ID1 (-0.154, 0.376, 0.099), ID2 (-0.103, 0.476, -0.027), ID3 (0.228, 0.036, -0.251), ID4 (0.330, 0.013, -0.263), ID5 (-0.114, 0.482, 0.014), ID6 (0.295, 0.084, -0.297), ID7 (0.262, 0.042, -0.304), ID8 (-0.051, 0.416, -0.306).

Table 7: Activity data set
Tabel 7: Aktiwiteitdatastel

ID	X	Y	Z
1	-0.154	0.376	0.099
2	-0.103	0.476	-0.027
3	0.228	0.036	-0.251
4	0.330	0.013	-0.263
5	-0.114	0.482	0.014
6	0.295	0.084	-0.297
7	0.262	0.042	-0.304
8	-0.051	0.416	-0.306

Question 1 *[7 marks]*

The K-means clustering approach should be applied to the data set provided in Table 8, with k = 2 and using Euclidean distance. Assume that the initial cluster centroids for the clusters C1 and C2 are c1 = <-0.235, 0.253, 0.438> and c2 = <0.232, 0.325, -0.159>. Calculate the distance between each data instance and each centroid. Use the results of the calculations to complete Table 8. Round your answers to three decimal figures.

Table 8: Activity data set – distance calculations
Tabel 8: Aktiwiteitdatastel – afstand berekeninge

ID	X	Y	Z	dist(**d**, c_1)	dist(**d**, c_2)
1	-0.154	0.376	0.099	0.370	0.467
2	-0.103	0.476	-0.027	0.532	0.390
3	0.228	0.036	-0.251	0.858	0.303
4	0.330	0.013	-0.263	0.931	0.343
5	-0.114	0.482	0.014	0.497	0.417
6	0.295	0.084	-0.297	0.922	0.285
7	0.262	0.042	-0.304	0.918	0.319
8	-0.051	0.416	-0.306	0.784	0.332

{Original student answer}

Answer:

Using Euclidean distance dist(d,c) = sqrt((x-cx)² + (y-cy)² + (z-cz)²):

ID1 (-0.154,0.376,0.099): dist to c1 ≈ 0.370; dist to c2 ≈ 0.686

ID2 (-0.103,0.476,-0.027): dist to c1 ≈ 0.532; dist to c2 ≈ 0.618

ID3 (0.228,0.036,-0.251): dist to c1 ≈ 0.858; dist to c2 ≈ 0.303

ID4 (0.330,0.013,-0.263): dist to c1 ≈ 0.921; dist to c2 ≈ 0.343

ID5 (-0.114,0.482,0.014): dist to c1 ≈ 0.497; dist to c2 ≈ 0.628

ID6 (0.295,0.084,-0.297): dist to c1 ≈ 0.923; dist to c2 ≈ 0.285

ID7 (0.262,0.042,-0.304): dist to c1 ≈ 0.919; dist to c2 ≈ 0.319

ID8 (-0.051,0.416,-0.306): dist to c1 ≈ 0.760; dist to c2 ≈ 0.532

(Based on these distances: ID1, ID2 and ID5 are closer to c1, so are assigned to Cluster 1; ID3, ID4, ID6, ID7 and ID8 are closer to c2, so are assigned to Cluster 2.)

Question 2 *[3 marks]*

Calculate the centroid values of cluster 1 after the 1st iteration of the K-means algorithm. Show all your calculations.

Answer:

Based on the assignment above, Cluster 1 after iteration 1 contains ID1, ID2 and ID5.

New centroid X = (-0.154 + -0.103 + -0.114) / 3 = -0.371 / 3 ≈ -0.124

New centroid Y = (0.376 + 0.476 + 0.482) / 3 = 1.334 / 3 ≈ 0.445

New centroid Z = (0.099 + -0.027 + 0.014) / 3 = 0.086 / 3 ≈ 0.029

New centroid for Cluster 1 ≈ <-0.124, 0.445, 0.029>